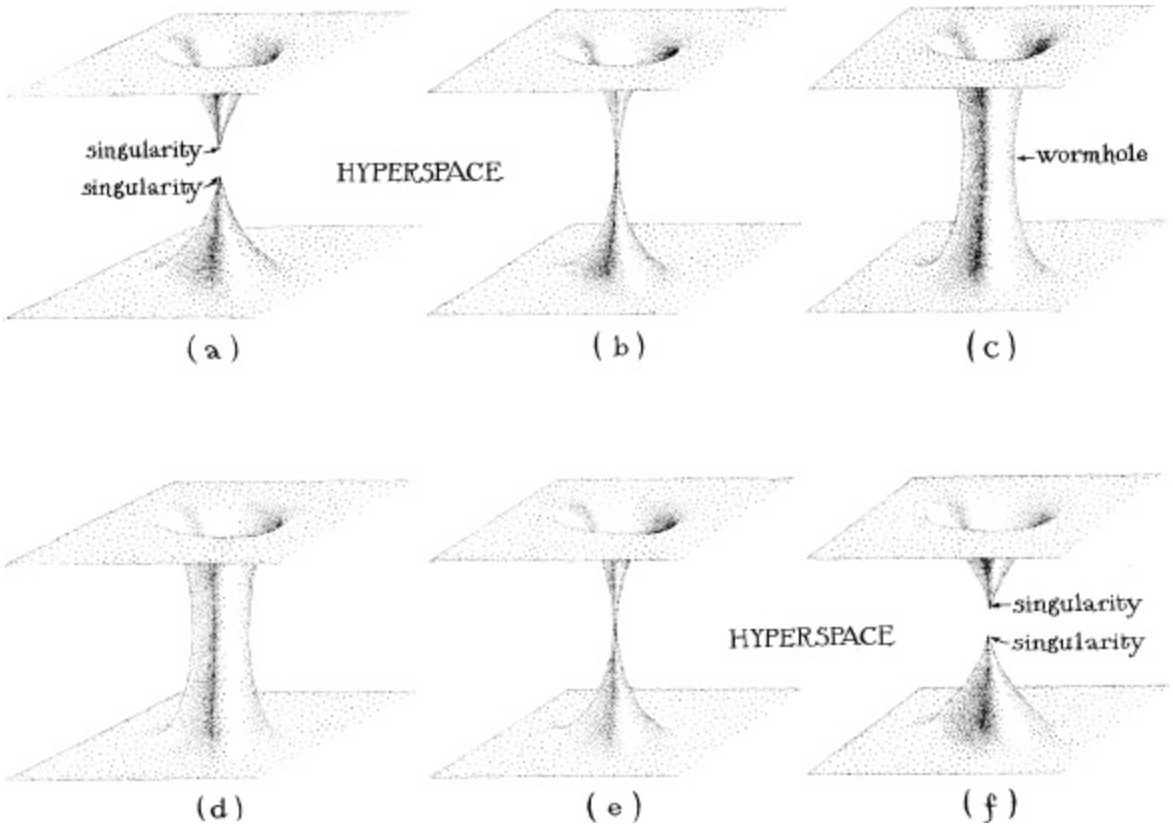


14.2 The evolution of a precisely spherical wormhole that has no material in its interior. (This evolution was discovered as a solution of Einstein's field equation in the mid-1950s by Martin Kruskal, a young associate of Wheeler's at Princeton University.) Initially (a) there is no wormhole; instead there is a singularity near Earth and one near Vega. Then, at some moment of time (b), the two singularities reach out through hyperspace, find each other, annihilate each other, and in the annihilation they create the wormhole. The wormhole grows in circumference (c), then begins to recontract (d), and pinches off (e), creating two singularities (f) similar to those in which the wormhole was born—but with one crucial exception. Each initial singularity (a) is like that of the big bang; time flows out of it, so it can give birth to something: the Universe in the case of the big bang, and the wormhole in this case. Each final singularity (f), by contrast, is like that of the big crunch (Chapter 13); time flows into it, so things get destroyed in it: the Universe in the case of the big crunch, and the wormhole in this case. Anything that tries to cross through the wormhole during its brief life gets caught in the pinch-off and, along with the wormhole itself, gets destroyed in the final singularities (f).

To make the calculations easy, I idealized the wormhole as precisely spherical (so in Figure 14.1, where one of our Universe's three dimensions is suppressed, it is precisely circular in cross section). Then, by two pages of calculations based on the Einstein field equation, I discovered three things:

First, *the only way to hold the wormhole open is to thread the wormhole with some sort of material that pushes the wormhole's walls apart,*